Introduction to Deep Learning in Computer Vision

Lecture 12: Video

Fall 2024

Aasa Feragen
afhar@dtu.dk

Today’s programme – and plan for final project presentations

- Lecture 13-14
- From 14-17:
  - Six sets of project presentations in building 358
  - TAs available in lecture hall
  - Start to work on Video project

- The video projects will be presented on the last day of teaching 4.12.2024
  - Here, we will run 4 parallel sessions of presentations with teachers and TAs to keep it short

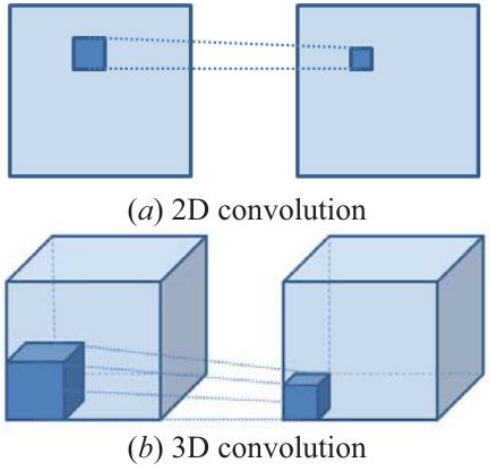
Today’s learning goals

You should be familiar with new challenges and task associated with video understanding, most notably:

- Aggregating per-frame models
- Late fusion
- Early fusion
- 3D CNNs

… and you should be ready to start a project implementing some of these techniques

So far? 2D images



- Natural extension: 3D images!
- How would you analyze those?
  - 3D CNNs!
- What's difficult with those?
  - Scalability – memory
- We won't do those – but a related case: Video
  - A video is a temporal sequence of images – a 4D tensor: T x 3 x H x W

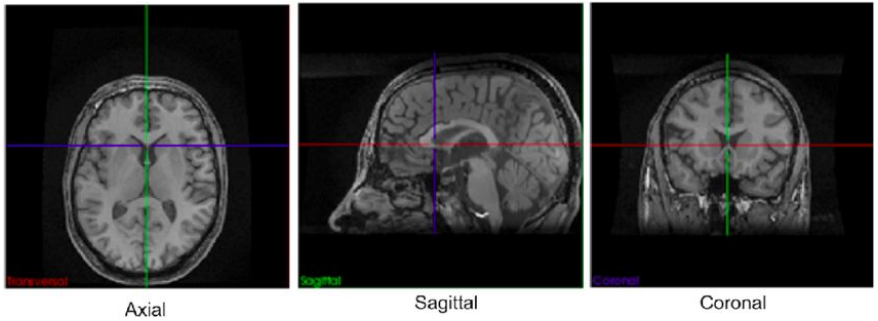


Figure from Mahmoudzadeh et al, Journal of Medical Imaging'14

Medical imaging example: Surgical video understanding

Ophthalmic surgical workflow understanding. Applications:

- Assessing quality of surgery – access to whole video
- Predicting next best course of action – access to previous frames

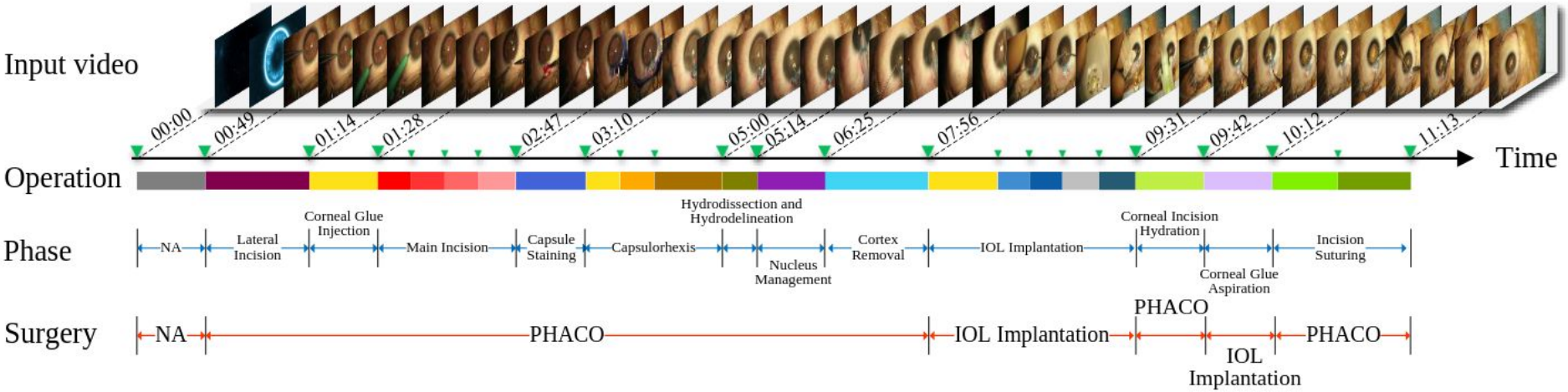


Figure from Hu et al, OphNet: A Large-Scale Video Benchmark for Ophthalmic Surgical Workflow Understanding, ECCV 2024

Real-world computer vision examples

- In the examples below: Realtime analysis, access only to previous frames

(a) Human-robot interaction

(b) Entertainment

(c) Autonomous driving car

Figure from Kong et al, IJCV 2022

What’s the challenge with video?

- **Scalability:**
  Frame rate of 24 frames per second (fps) gives 1440 images per minute – consider that in light of your batch size limitations
- Utilizing the temporal dimension
  - Today: simple approaches
  - Coming weeks – more direct modeling
- New tasks!

Figure from Hu et al, OphNet: A Large-Scale Video Benchmark for Ophthalmic Surgical Workflow Understanding, ECCV 2024

Our running example task: Video classification

Detecting **actions**, not **objects**

Input:
T x 3 x H x W

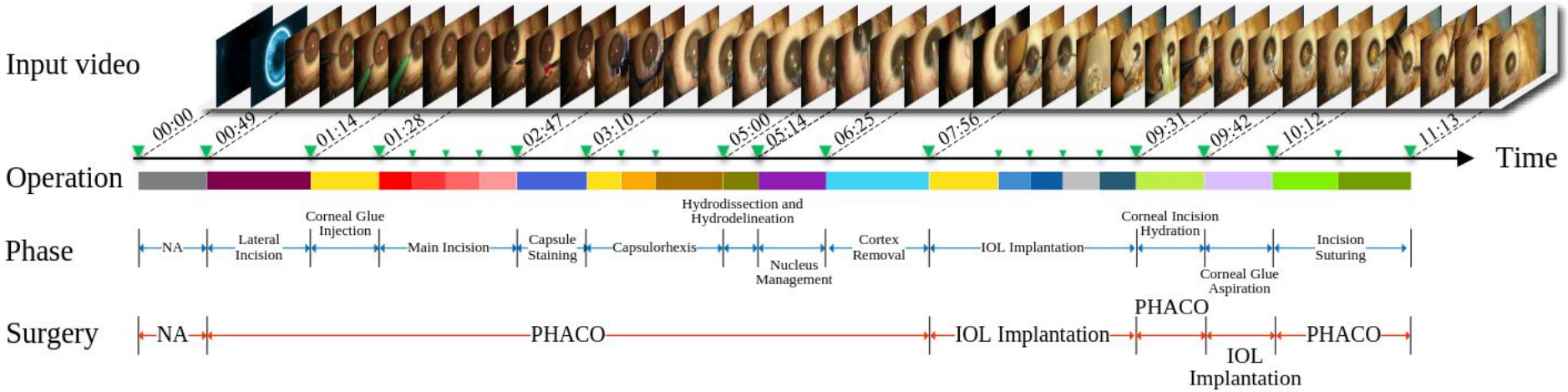
Video classification model

Output:
Blinking
Yawning
Smiling
Talking
Whistling

Illustration based on the Talking Face dataset
(https://personalpages.manchester.ac.uk/staff/timothy.f.cootes/data/talking_face/talking_face.html)

New task: Action localization

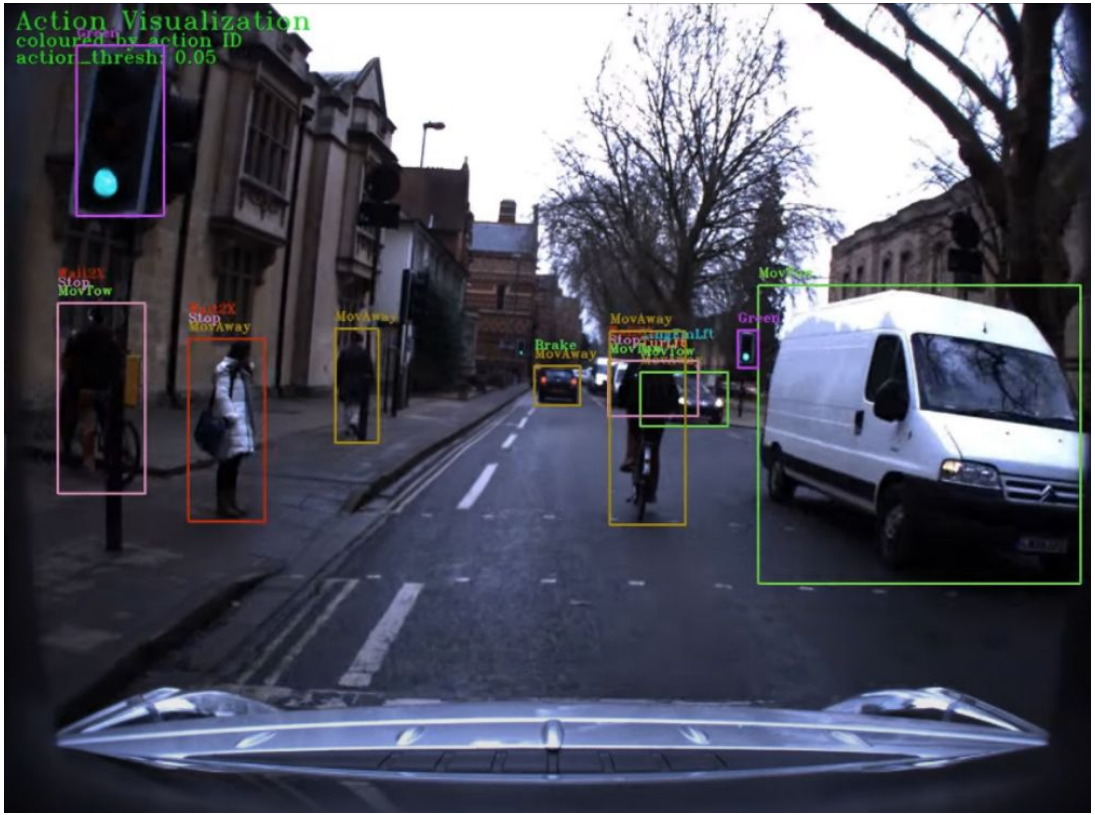
- Temporal location + classification of actions



Hu et al, OphNet: A Large-Scale Video Benchmark for Ophthalmic Surgical Workflow Understanding, ECCV 2024

New task: Spatiotemporal action localization

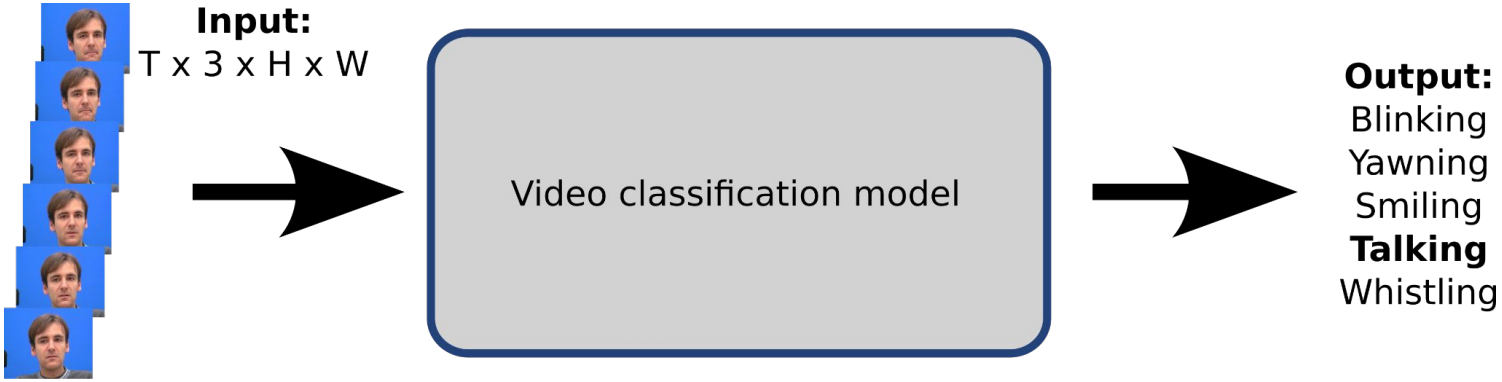
- Locating action in both space and time – and classifying it



Our running example: Video classification

Tackling scalability?

- Downsample in time
- Downsample in space
- Analyze short clips
- Needless to say, both come with a performance cost



One solution: Clip-wise analysis

Raw video: Long, high FPS

Training: Train model to classify short **clips** with low FPS

Testing: Run model on different clips, average predictions

Slide credit: Justin Johnson

Figure from https://cs231n.stanford.edu/slides/2023/lecture_10.pdf

Simple special case: Classify single frames and aggregate

→ 2D CNN → Frowning

→ 2D CNN → Frowning

→ 2D CNN → Frowning

→ 2D CNN → Talking

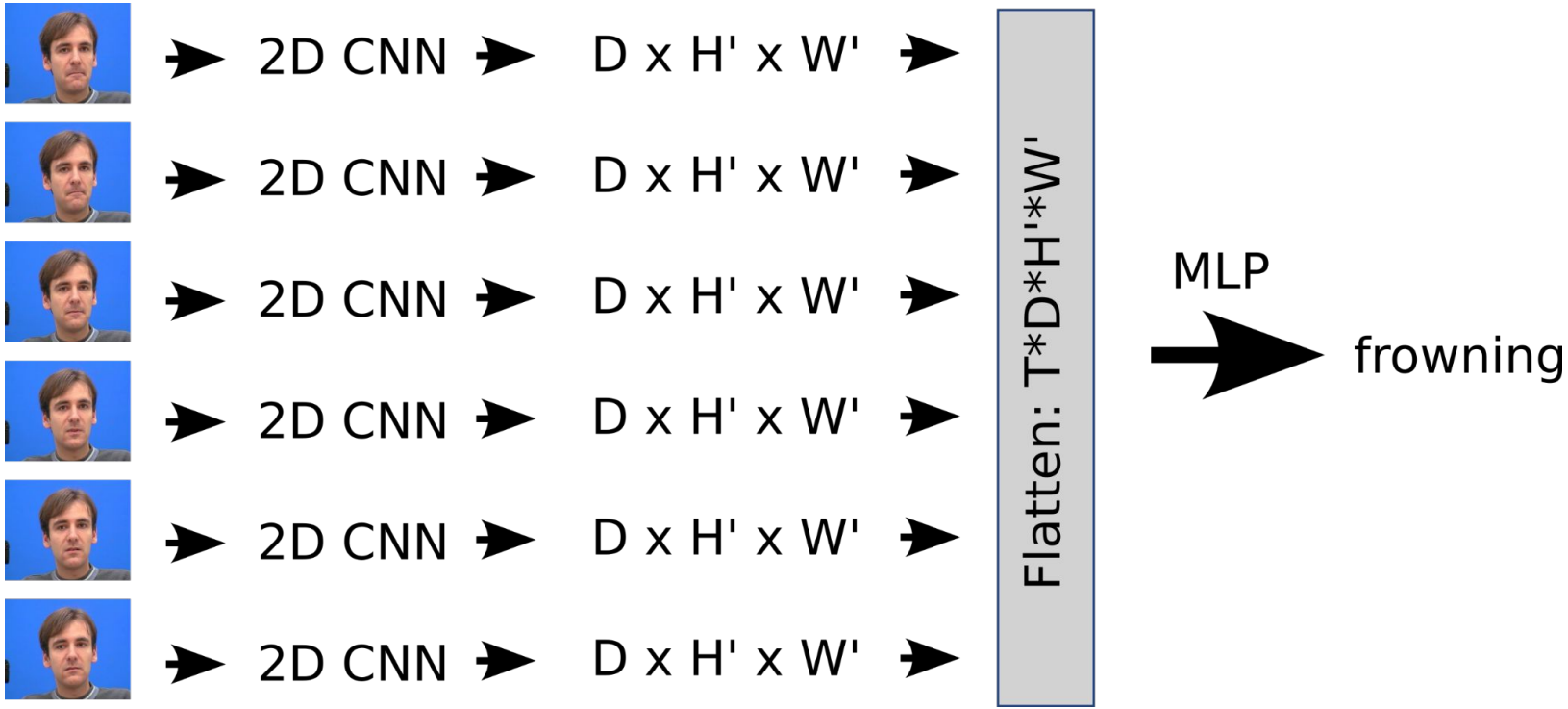
→ 2D CNN → Talking

→ 2D CNN → Frowning

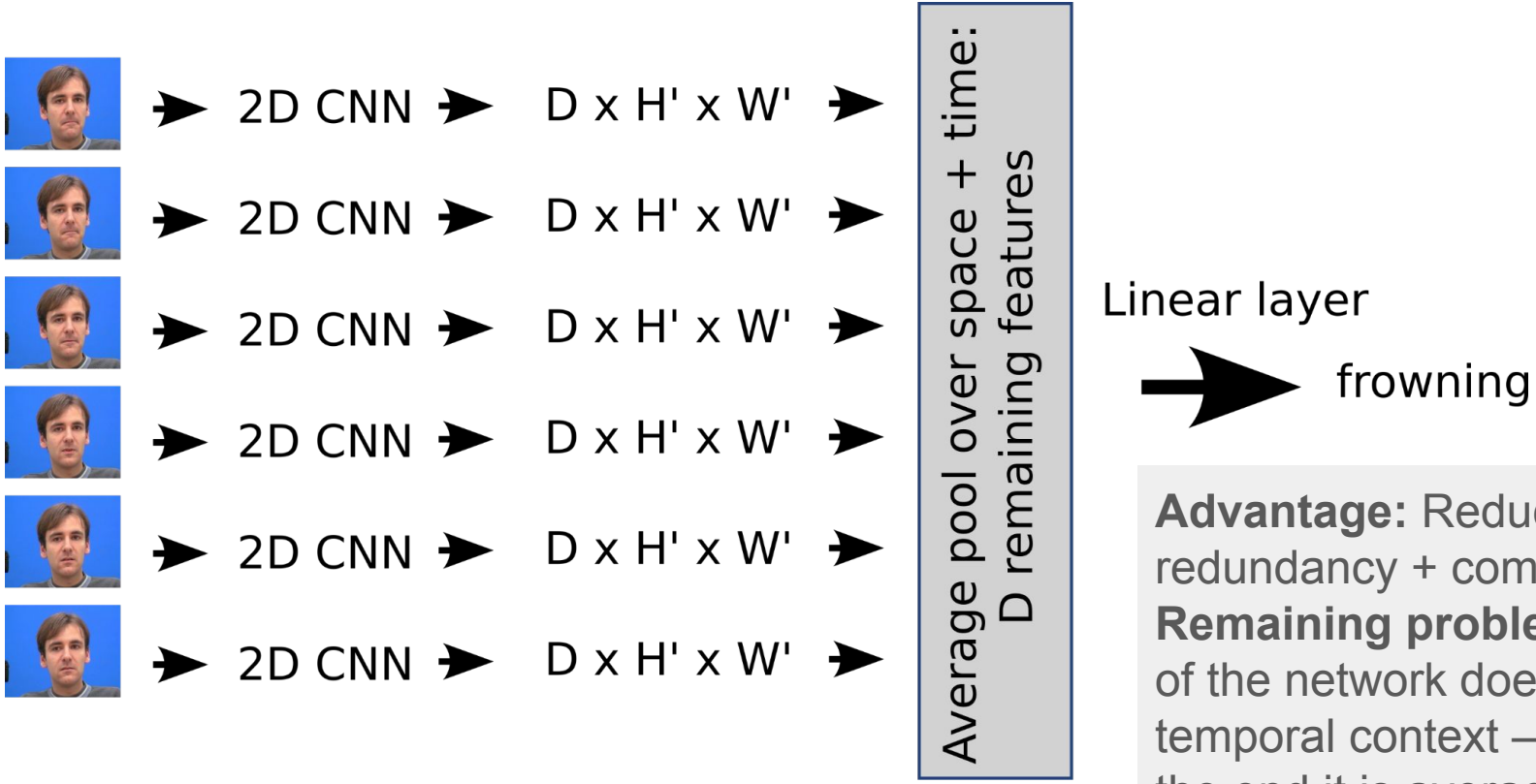
Aggregate,
e.g. by averaging:
Frowning

Train CNN on independent frames, aggregate predictions – strong baseline

Late fusion 1: Aggregate frames with an MLP

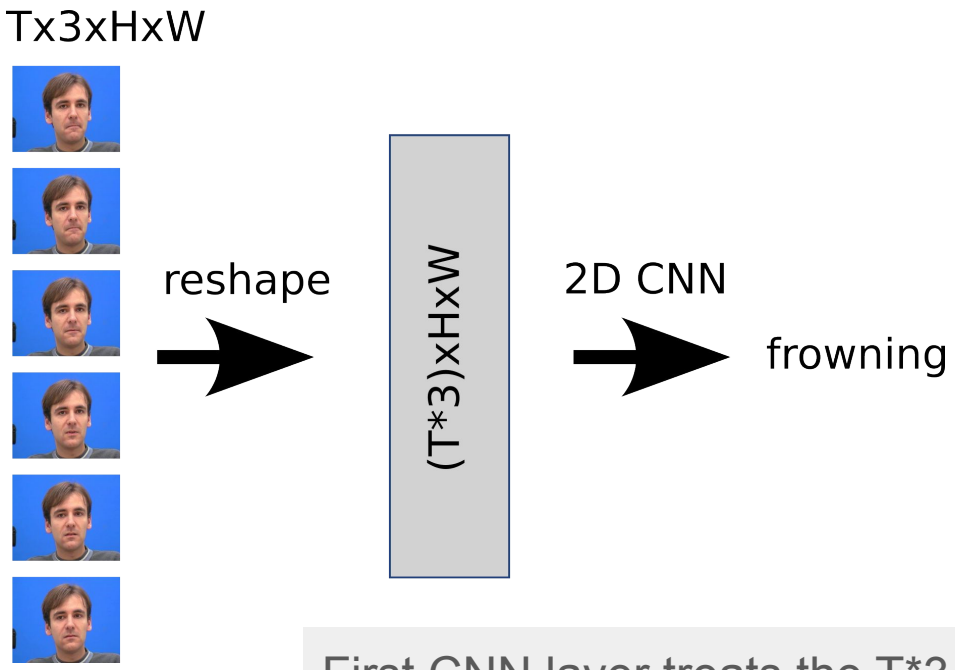


Late fusion 2: With pooling



Advantage: Reduce some redundancy + complexity.
Remaining problem: Most of the network does not have temporal context – and at the end it is averaged out.

Early fusion



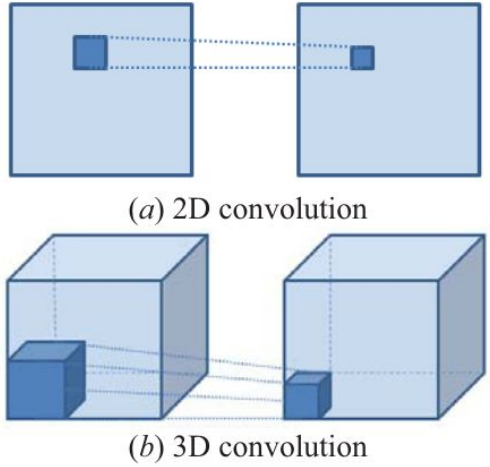
First CNN layer treats the T*3 time + RGB channels as feature channels, transformed to D abstract feature channels

3D CNNs

3xTxHxW

3D CNN

frowning



3D CNN consisting of 3D convolutional layers with 3D kernels, and 3D pooling layers.

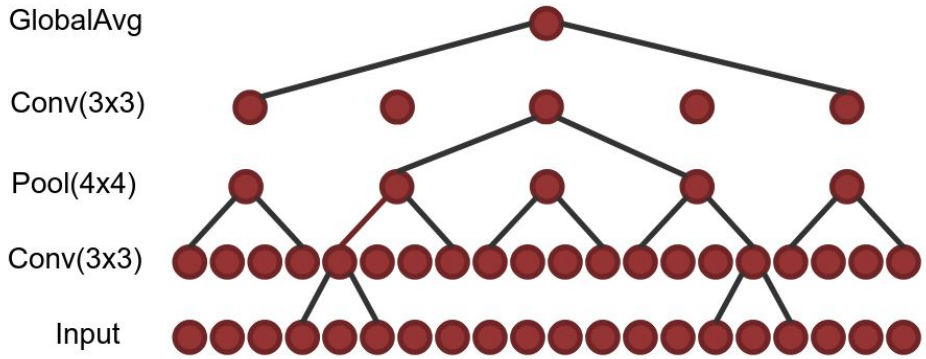
Convolution figure from Fan et al, FADS'17

How are these approaches different?

Late Fusion

Layer	Size (C x T x H x W)	Receptive Field (T x H x W)
Input	3 x 20 x 64 x 64	
Conv2D(3x3, 3->12)	12 x 20 x 64 x 64	1 x 3 x 3
Pool2D(4x4)	12 x 20 x 16 x 16	1 x 6 x 6
Conv2D(3x3, 12->24)	24 x 20 x 16 x 16	1 x 14 x 14
GlobalAvgPool	24 x 1 x 1 x 1	20 x 64 x 64

Spatial receptive field expands gradually
Time receptive field all in one go at the end



Example + illustration from https://cs231n.stanford.edu/slides/2023/lecture_10.pdf

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Early Fusion	Input	3 x 20 x 64 x 64	
	Conv2D(3x3, 3*20->12)	12 x 64 x 64	20 x 3 x 3
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Spatial receptive field expands gradually
Time receptive field all in one go from the beginning

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	GlobalAvgPool	24 x 1 x 1	20 x 64 x 64
3D CNN	Input	3 x 20 x 64 x 64	
	Conv3D(3x3x3, 3->12)	12 x 20 x 64 x 64	3 x 3 x 3
	Pool3D(4x4x4)	12 x 5 x 16 x 16	6 x 6 x 6
	Conv3D(3x3x3, 12->24)	24 x 5 x 16 x 16	14 x 14 x 14
	GlobalAvgPool	24 x 1 x 1	20 x 64 x 64

Spatial receptive field expands gradually
Time receptive field all in one go at the end

Spatial receptive field expands gradually
Time receptive field all in one go from the beginning

Spatial and time receptive fields both expand gradually

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